Kevin Guan

Curriculum Vitae

Manalapan, New Jersey — kevin0guan@gmail.com — www.linkedin.com/in/kevin-j-guan/

EDUCATION

Princeton University
2023-2027
Bachelor of Arts in Mathematics
Minors in Linguistics, Music, and Dance Studies
GPA: 3.83
Relevant Coursework: Functional Analysis, Molecular Modeling Methods, Reinforcement Learning, fMRI Decoding, Probabilistic Methods in Combinatorics, Probability in High Dimensions, Probability Theory, Information Theory, Theory of Computation, Graph Learning, Complex Analysis, Quantum Mechanics, Machine Learning

TECHNICAL EXPERIENCE

Princeton Laboratory for Artificial Intelligence
Princeton, New Jersey
Undergraduate Researcher, June 2025–Present
Advisors: Dr. Happy Buzaaba, Prof. Christiane Fellbaum
Research on syntactic parsing for low-resource African languages. First-authored a study comparing recurrent and transformer-based parsers across ten typologically diverse languages; used mixed-effects regression to show that LSTM parsers outperform pretrained transformers below a crossover threshold, with morphological complexity as a significant secondary predictor of transformer disadvantage. Contributed to AfriSUD, a nine-language Surface-Syntactic UD treebank benchmark, by implementing dependency-tree validation and building the model evaluation suite. Established supervised parsing baselines and conducted zero-/few-shot cross-lingual transfer experiments with Stanza and multilingual encoder models for POS tagging and dependency parsing.

Fields Institute for Research in Mathematical Sciences
Toronto, Canada
Research Intern, June 2026-July 2026
Advisors: Dr. Divya Sharma, Dr. Jude Kong
Developed a physics-informed neural network (PINN) to replace classical ODE epidemic models with a data-driven surrogate, recovering epidemiological parameters via inverse-problem estimation. Introduced Monte Carlo dropout as a learned alternative to SDE process noise for generating stochastic trajectories. Trained PPO agents on the learned simulator to study reinforcement-learned intervention policies under different pathogens, model structures, and geographies with cross-region validation against real government policy responses.

Complexity Science Hub
Vienna, Austria
Research Intern, June 2025–August 2025
Advisor: Dr. Eddie Lee
Constructed a large-scale dynamic financial flow network from 51M+ political transaction records, using fuzzy matching and LLM-based entity resolution to enable cross-election graph comparison via trajectory analysis. Applied structural and spectral graph analysis to characterize network topologies, identify higher-order dynamics, and uncover differences in partisan funding strategies. Trained predictive models of financial behavior—including a high-dimensional recursive linear model and a random forest—and applied SHAP-value attribution to identify key structural drivers of political spending.

Center for Statistics and Machine Learning at Princeton University
Princeton, New Jersey
Research Assistant, February 2024–July 2024
Constructed a demographic dataset for ranked choice voting elections by integrating GIS shapefiles and American Community Survey (ACS) microdata using Python. Applied statistical imputation techniques to reconstruct missing observations and aggregated Census tract-level data to custom geographic units. Assessed and documented data quality through systematic evaluation of coverage and consistency metrics, producing a formal methodology report. Developed a Flask-based annotation interface for labeling gubernatorial speech data to support supervised training of a large language model.

Bloomberg L.P.
New York City, New York
Software Engineering Intern, July 2022–August 2024
Completed three internships across software engineering and machine learning. Developed an OCR pipeline and fine-tuned a transformer model to extract structured field data from unstructured municipal bond documents, evaluating model iterations using token-level metrics to guide optimization. Built a Python framework to detect redundant data across 4,000+ SQL tables, incorporating NLP-based column name similarity scoring and a decision tree classifier trained on linguistic features to assess probable column equivalence. Migrated a high-traffic terminal feature (150,000+ daily users) from JavaScript to TypeScript, applying object-oriented design principles.

RESEARCH MANUSCRIPTS

Evaluating Architectures on High and Low-Resource Languages
Kevin Guan, Happy Buzaaba, and, Christiane Fellbaum
Accepted to *Language Technologies for Low-Resource Languages (LaTeLL) 2026*
Preprint: arXiv:2605.02608

AfriSUD: A Dependency Treebank Collection for Evaluating Models on African Languages
Happy Buzaaba, Cheikh Mouhamadou Bamba Dione, David Ifeoluwa Adelani, Sylvain Kahane, Kim Gerdes, Bruno Guillaume, **Kevin Guan**, et al.
Submitted to *Empirical Methods in Natural Language Processing (EMNLP) 2026*
Preprint: arXiv:2606.12708

Recovering the Temporal Structure of Drifting Data from Model Weights
Kevin Guan
Submitted to *Transactions on Machine Learning Research (TMLR)*
Preprint: arXiv:2607.27482

HONORS, AWARDS, AND GRANTS

High Meadows Summer Internship Grant (2026)
Awarded $7,200 by High Meadows Environmental Institute
Competitive grant supporting a full-time summer research internship in mathematical epidemiology at the Fields Institute, awarded for work at the intersection of computational modeling and environmental or public health.

International Internship Program Award (2025)
Awarded $4,640 by Princeton Office of International Programs
Competitive merit-based award granted to fund a 10-week summer internship abroad. Selected from a university-wide applicant pool to conduct independent research at the Complexity Science Hub in Vienna, Austria, an institute dedicated to the quantitative study of complex systems across science, economics, and society.

Outstanding Work by a Sophomore: Dance (2025)
Awarded $125 by Lewis Center for the Arts
Faculty-nominated award recognizing outstanding achievement within the Program in Dance.

MathWorks Math Modeling Challenge: Third Place Award (2023)
Awarded $10,000 by Society for Industrial and Applied Mathematics
Co-authored a 14-hour team paper modeling e-bike adoption and its transportation impacts in the US and UK, placing third of 650 teams. Led two of three modeling components: (1) fit a Bass diffusion model to historical adoption data via nonlinear least-squares regression to forecast 2- and 5-year e-bike sales in the US and UK; (2) applied a random forest algorithm to rank the causal significance of factors—including urban population, electricity prices, and disposable income—driving e-bike adoption. Conducted sensitivity analysis on both models.

MathWorks Math Modeling Challenge: Finalist and Technical Computing Award (2022)
Awarded $7,000 by Society for Industrial and Applied Mathematics
Co-authored a 14-hour team paper forecasting remote-work adoption and its economic effects across five US and UK cities (Finalist, top 6 of 612 teams; Technical Computing Award Runner-Up). Developed an individual worker-classification model by trained an interpretable RuleFit classifier on American Time Use Survey data to predict work-from-home decisions from demographic factors including age, occupation, education, and household composition. Evaluated performance via confusion matrices and statistical metrics.